Lab 4 Rubric: Newton's Second Law

Out of 100 points. This is the rubric on the LB4-M3 assignment in Brightspace. Lab 4 page · Word copy · PDF copy

Use this rubric as your writing guide: the criteria follow the lab report template top to bottom. Full-credit descriptors say exactly what a complete answer contains, so read the Complete column before you start writing, not after. Note 6 is where the physics for the Introduction and the Experimental section comes from — bringing it into the report is expected, and is marked below.

Grammar, spelling, and formatting problems are assessed 1 point per occurrence. Deadline: a report submitted by the deadline may be revised and resubmitted up to two more times. A report that arrives after the deadline but before grades are posted is graded as on time but cannot be revised; one that arrives after grades are posted is late, may take a penalty, and gets no detailed feedback.

The phrase “human error” earns no credit anywhere in the report. A percent error under 10% may be attributed to ordinary experimental uncertainty, provided you say so explicitly and support it with the uncertainty on your fitted slope. Above 10%, named and quantified error sources are required.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Heading	3	**3** Names, course and section, lab title and date, all present and correctly formatted.	**2** One item missing or misformatted.	**2** Template placeholders or brackets left in the heading, or the section number missing.	**1** Several items missing.	**0** No heading block.
Abstract	7	**7** Four sentences or fewer. States what was measured, gives the experimental total mass with its uncertainty and the percent error, and says whether Newton's second law was verified. Nothing else.	**6** All content present but runs long or includes method detail.	**4** Reports the experimental mass or the percent error but not both, or gives the mass with no uncertainty.	**2** Describes the lab but reports no numbers.	**0** No abstract.
Introduction	10	**10** Derives a = F_g,H / M_T line by line from the two free-body equations, eliminating the tension; rearranges it to a = (1/M_T)F and says in words why that makes the slope of a against F a mass; every variable defined; states a hypothesis that can later be confirmed or rejected.	**8** Derivation present but a step, the rearrangement, or a variable definition is missing.	**6** The derivation is sketched rather than worked line by line, or the rearrangement to a = (1/M_T)F is shown without saying in words why that makes the slope a mass.	**3** The equation is quoted rather than derived, or there is no hypothesis.	**0** No introduction.
Experimental: Materials & Setup	8	**8** Apparatus described in prose; a labelled sketch of the half Atwood setup redrawn by you, AND a free-body diagram for each body, with the positive direction chosen for each body stated explicitly.	**7** Sketch and both free-body diagrams present and labelled, but one force label or axis choice is unclear.	**5** Only one body has a free-body diagram, or the sign convention is not stated for either body.	**3** Bare equipment list, or only one of sketch and free-body diagrams.	**0** No materials or setup description.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Experimental: Procedure	8	**8** Written in own words as prose, reproducible, and EXPLAINS THE CONSTANT-TOTAL-MASS STRATEGY: that mass was moved from the cart to the hanger rather than added, and why that is what leaves the driving force as the only variable under test. States how each acceleration was obtained from the position data.	**7** Reproducible and in the student's own words, but one detail is thin.	**5** Reproducible, but the constant-total-mass strategy is stated without being explained.	**3** Copied or closely paraphrased from the handout.	**0** No procedure.
Results: Data Tables	12	**12** Table of all five runs carrying hanging mass, driving force, the fit window used, the quadratic coefficient, the acceleration and the uncertainty the fit reported. Every table numbered, captioned, with units and consistent significant digits, and referenced in the text.	**10** Table present; minor issues with captions, units or significant digits, or the fit windows are omitted.	**7** The run table is present but omits the fit windows or the fit uncertainties, or covers fewer than five runs.	**4** Fewer than five runs tabulated, or the driving-force column missing.	**0** No data tables.
Results: Graphs	9	**9** Acceleration against driving force, with the fitted straight line, axes labelled with units, numbered and captioned and described in the text; AND at least one position-versus-time graph shown with its fitted parabola, so a reader can see which points were used.	**7** Both graphs present and correctly plotted; a caption or the in-text description is incomplete.	**5** The a-versus-F graph is complete, but no position-versus-time graph is shown.	**3** Points joined dot-to-dot, or no fitted line.	**0** No graph.
Results: Sample Calculations	13	**13** One worked sample of each type, with units at every step: driving force from the hanging mass, acceleration from the quadratic coefficient, total mass from the slope, propagation of the slope's uncertainty into the mass, and percent error.	**11** All types present with minor unit or rounding slips.	**8** Three or four of the five types worked; one or two missing.	**4** Some types missing, or units given only in the final answer.	**0** No sample calculations.
Discussion	15	**15** Opens with the experimental mass ± uncertainty; addresses BOTH verification criteria by name — that acceleration rises in proportion to force, AND that the slope equals 1/M_T; discusses the intercept rather than ignoring it; explains the DIRECTION of the discrepancy and not only its size; states whether the hypothesis is confirmed and argues it with the 10% rule.	**12** Most of the above; one element thin or missing — most often the intercept, or the direction of the discrepancy.	**9** Opens with a result and interprets it, but two or three of the required moves are absent — one of the two verification criteria, the intercept, the direction of the discrepancy, or the 10% argument.	**5** Restates the results without interpreting them, or treats a straight line on its own as proof of the second law.	**0** No discussion.
Additional Questions	5	**5** All three Additional Questions (AQ1–AQ3) answered in paragraphs, in a dedicated section.	**4** One of the three answered thinly.	**3** Answered inside another section rather than under their own heading.	**2** Answered as a bare list, or one of the three skipped.	**0** Not answered.
Conclusion	5	**5** Three to four sentences: numerical result with percent error, a definitive verified / not-verified statement, and one concrete improvement to the measurement.	**4** Present but missing the numbers or the verdict.	**3** Present with the numbers but no definitive verified / not-verified verdict, or no improvement named.	**2** Present but vague, or a repeat of the discussion.	**0** No conclusion.
References	5	**5** At least one source beyond the textbook; Note 6 and this handout both count. Author, title and date given for each.	**4** Multiple references but formatting incomplete.	**3** Textbook and handout only, properly formatted.	**2** Handout only.	**0** No references.
Total	100					

Overall: Complete 90+ · Nearly there 75+ · Needs work 50+ · Missing below 50.

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